THAKUR COLLEGE OF SCIENCE AND COMMERCE

Affiliated to University of Mumbai 2026 – 2027

PRACTICAL JOURNAL OF
RESEARCH IN COMPUTING

SUBJECT GUIDE
Mrs. DHANASHREE CHAUDHARI
(Asst. Professor)

SUBMITTED BY:
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ROLL NO: 264928

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR QUALIFYING M.Sc.DS PART I
(SEMESTER I EXAMINATION)

Thakur Educational Trust's (Regd.)
THAKUR COLLEGE OF SCIENCE & COMMERCE
AUTONOMOUS COLLEGE, PERMANENTLY AFFILIATED TO UNIVERSITY OF MUMBAI
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CERTIFICATE OF APPROVAL

This is to certify that **Mr. Vinay Sanjay Panchal** student of "Master of Science (DATA SCIENCE)" of "Thakur College of Science and Commerce". Roll No. **264928** has successfully completed and submitted the Practical & Assignment entitled "**Research in Computing**" in the partial fulfilment as per the syllabus defined by the University of Mumbai in the academic year 2026-2027.

It is further certified that the student has completed all the required phases of the practical & assignment.

HEAD OF DEPARTMENT

PROFESSOR INCHARGE

EXTERNAL EXAMINER

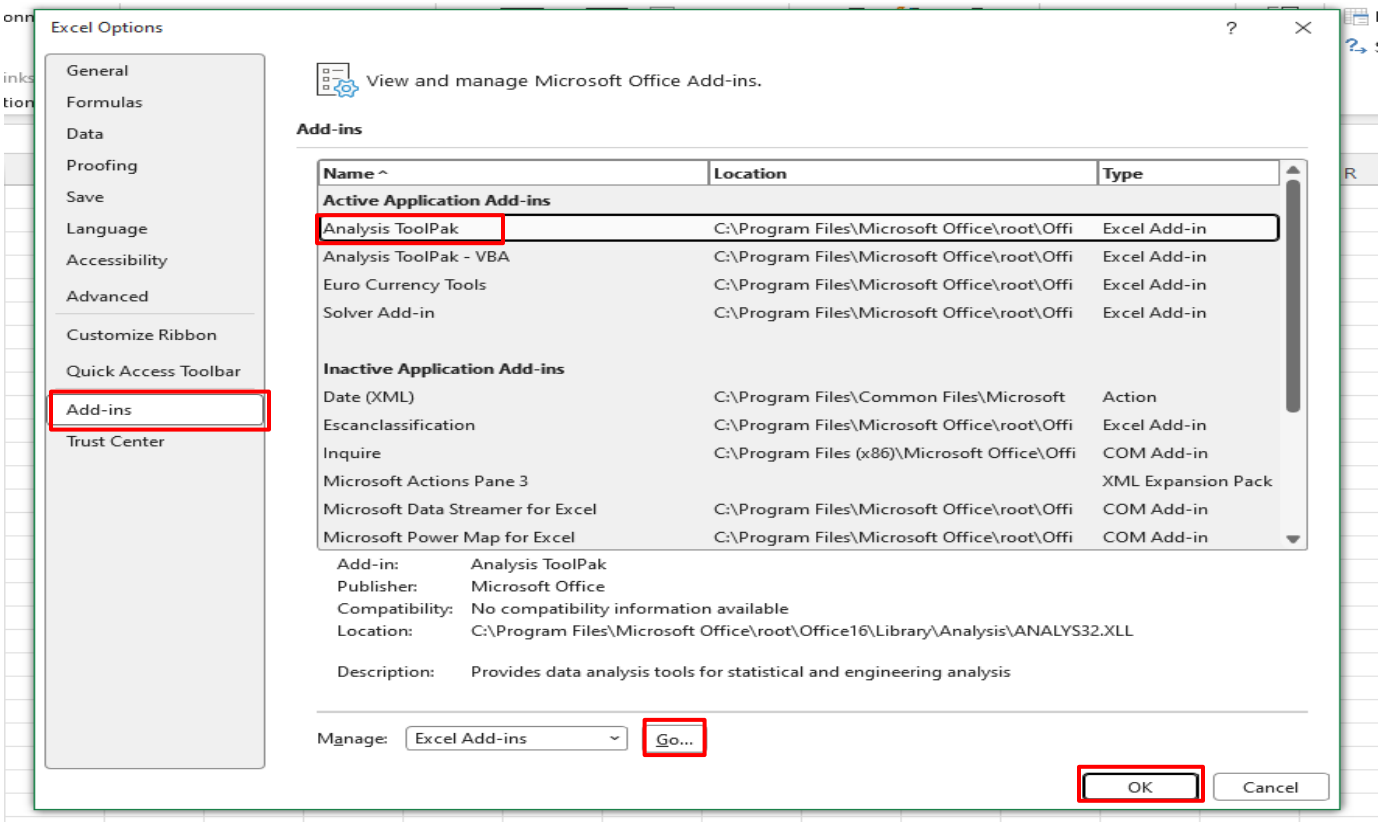
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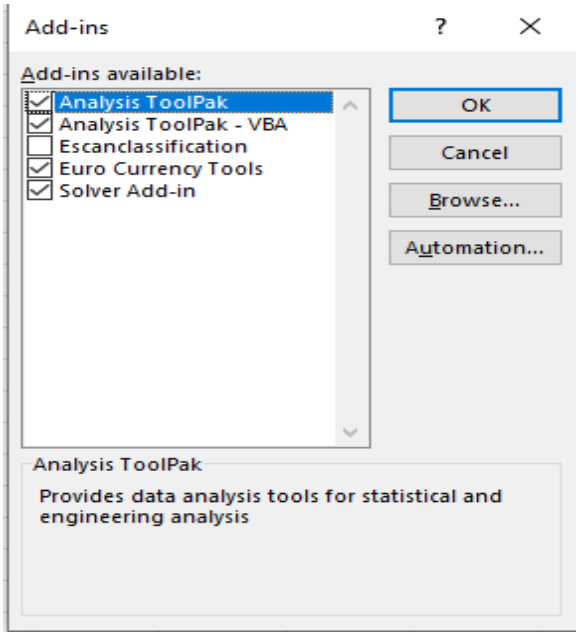
Practical 1

<u>Aim:-</u> Obtaining descriptive statistics of data using Excel.

Step 1:- Go to file -> Options -> Add-Ins -> Select Analysis ToolPak -> Press Ok



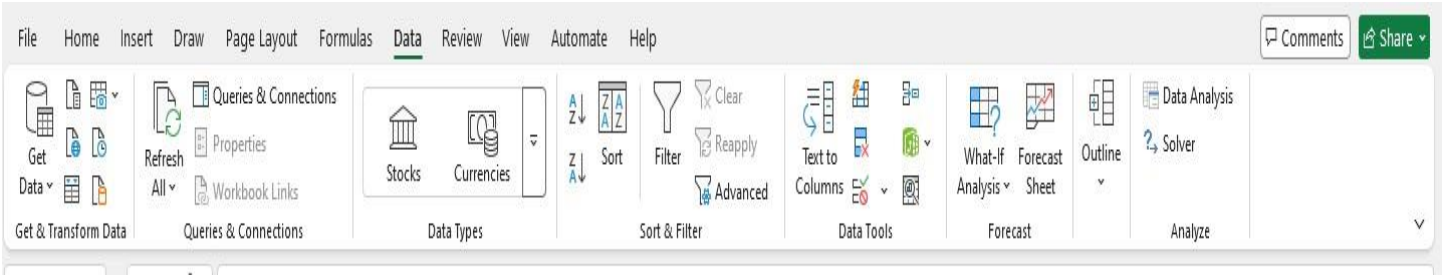
Step 2:- Select all four options in add-ins and click ok



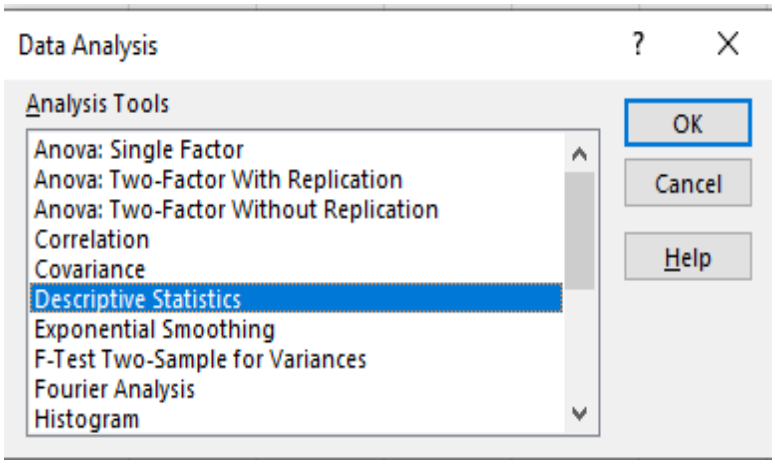
Step 3:- Enter the data manually

Product A	Product B	Product C
165	137	283
222	202	288
172	104	333
176	207	387
245	151	274
214	203	398
156	173	284
163	267	298
156	184	283
241	199	302
198	189	384
175	163	284
185	173	299
284	152	284
280	188	363

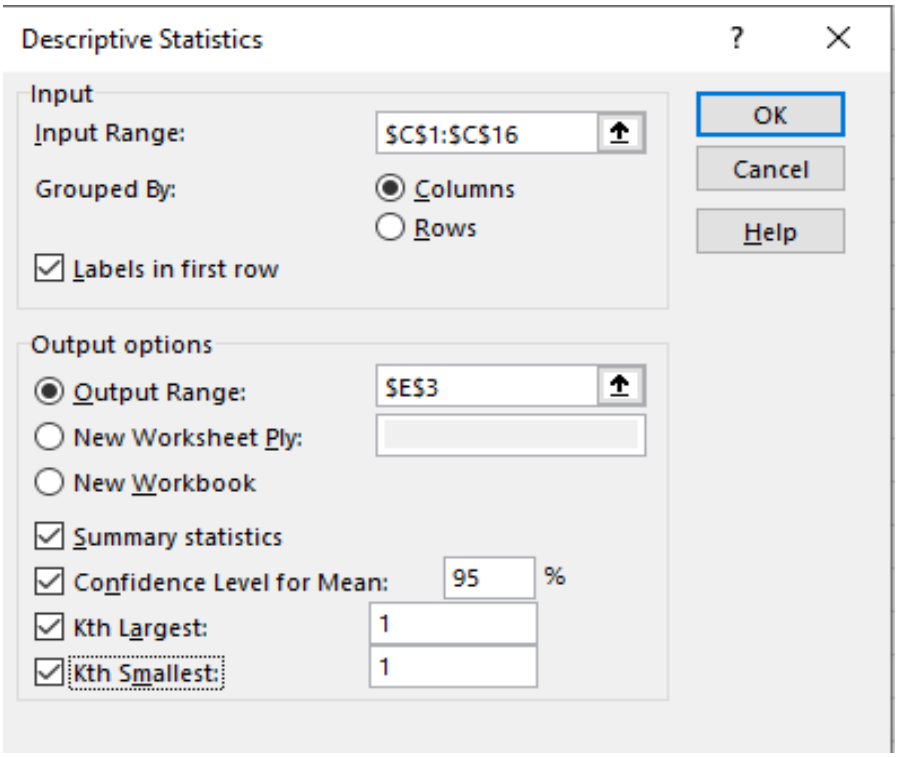
Step 4:- Click on “Data”, then select “Data Analysis” (top option at the right side)



Step 5:- Select “Descriptive Statistics” , then Click Ok



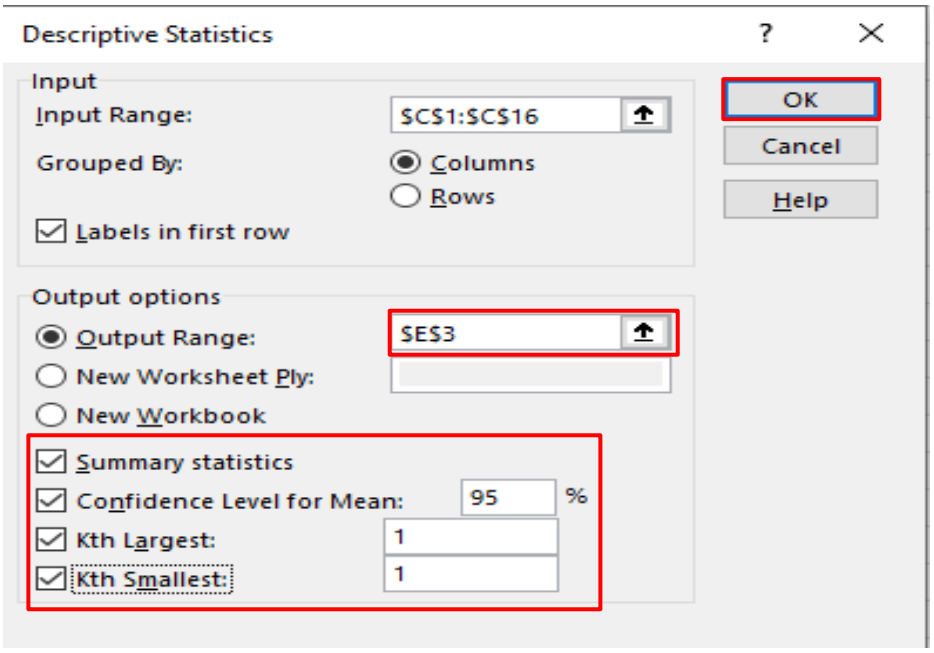
Step 6:- Provide input rate, (If labels are also selected then click on the check below as “Labels in first row”



Step 7:- Select the data range from the excel worksheet

Product A	Product B	Product C
165	137	283
222	202	288
172	104	333
176	207	387
245	151	274
214	203	398
156	173	284
163	267	298
156	184	283
241	199	302
198	189	384
175	163	284
185	173	299
284	152	284
280	188	363

Also, don’t forget to provide output range and select option like summary statistics, confidence level, Kth largest, Kth smallest, then lastly Click on Ok.



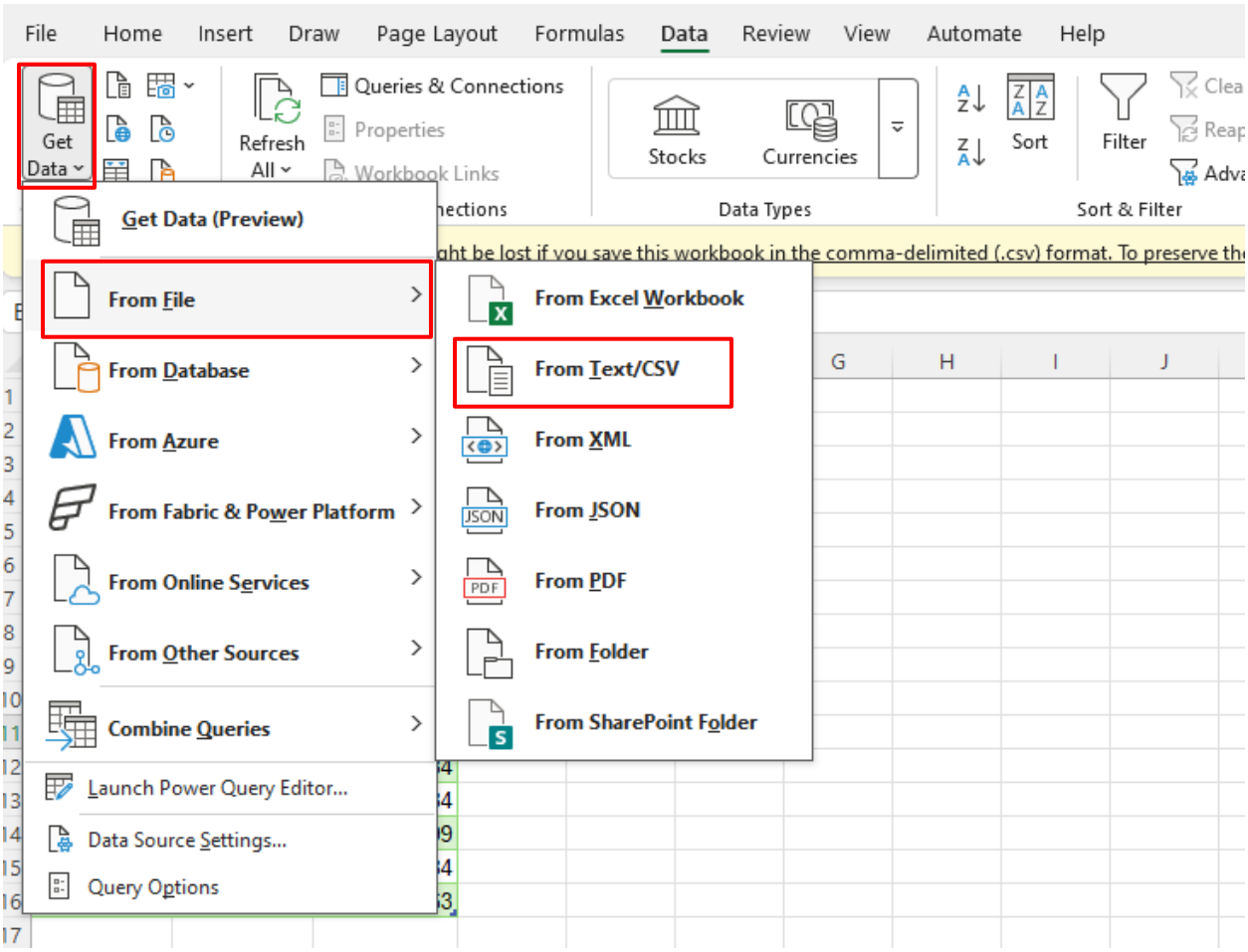
Output:-

	A	B	C	D	E	F
1	Product A	Product B	Product C			
2	165	137	283			
3	222	202	288		*Product C*	
4	172	104	333			
5	176	207	387		Mean	316.2667
6	245	151	274		Standard I	11.44197
7	214	203	398		Median	298
8	156	173	284		Mode	284
9	163	267	298		Standard I	44.31457
10	156	184	283		Sample Va	1963.781
11	241	199	302		Kurtosis	-0.76788
12	198	189	384		Skewness	0.976184
13	175	163	284		Range	124
14	185	173	299		Minimum	274
15	284	152	284		Maximum	398
16	280	188	363		Sum	4744
17					Count	15
18					Largest(1)	398
19					Smallest(1	274
20					Confidenc	24.54059
21						

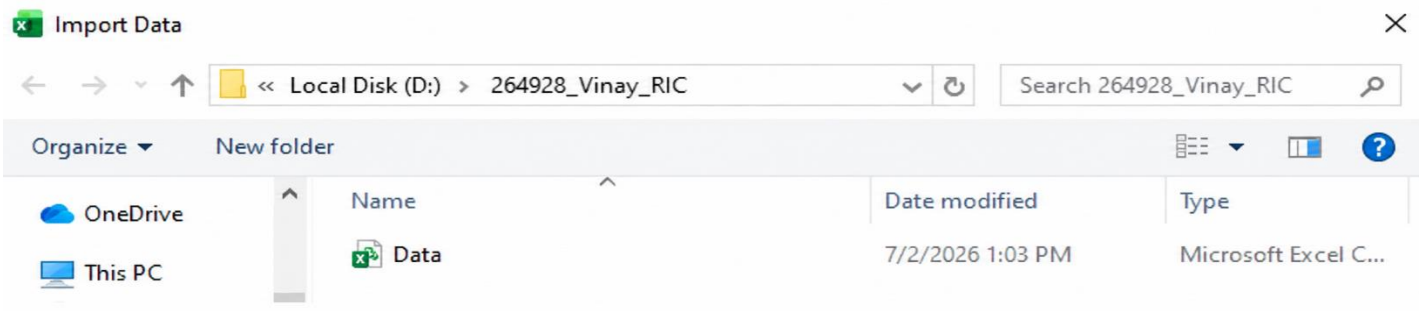
Practical 2

<u>Aim:-</u> Import data from excel and CSV to Excel

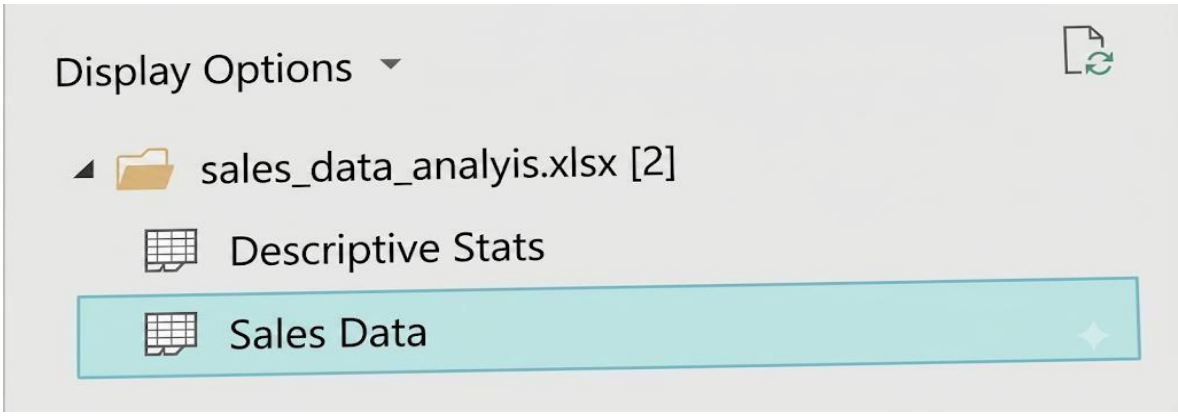
Step 1:- Open new "Blank Workbook" in Excel, then go to data -> Click on Existing Connection, then select "Browse for more" **OR Click Get Data -> From File-> From Text/CSV**(for csv files).



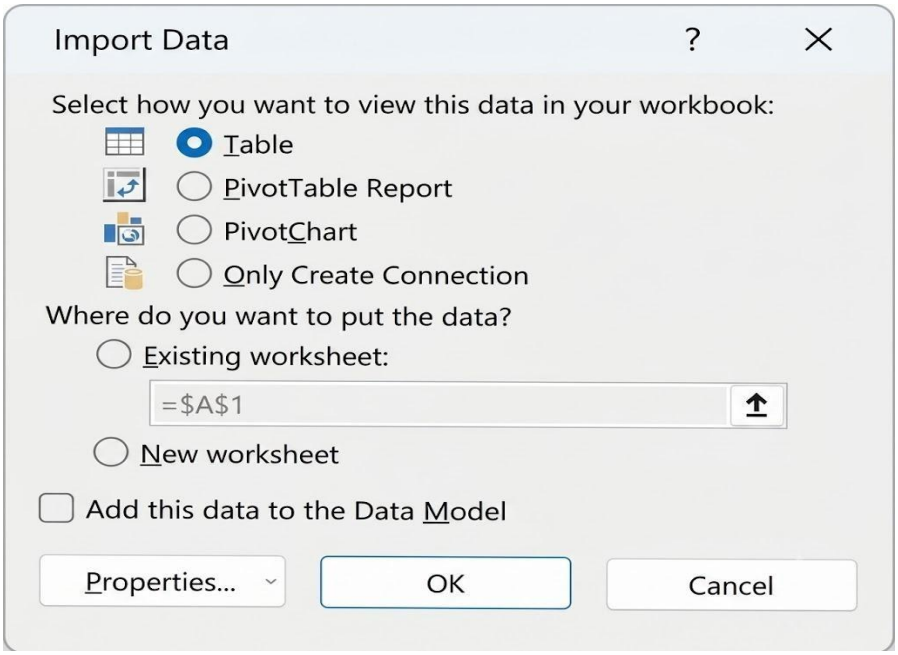
Step 2:- Now locate and select your file.



Step 3:- Select the sheet which you want to import and Click Ok



Step 4:- This dialog box ask us about how we want to view this data in our workbook, we don't have to do anything here, just Click Ok.



Output:-

	A	B	C
1	Product A	Product B	Product C
2	165	137	283
3	222	202	288
4	172	104	333
5	176	207	387
6	245	151	274
7	214	203	398
8	156	173	284
9	163	267	298
10	156	184	283
11	241	199	302
12	198	189	384
13	175	163	284
14	185	173	299
15	284	152	284
16	280	188	363

Practical 3(a)

Aim:- Design a survey form for a given case study, collect the primary data and analyze it.

Case Study:-

An organization want to survey users of its online payment app (such as Z-fast Pay) in order to improve the product by learning about their needs.

- Questions asked by the company to their customers are as follows:-

1. Demographic Information:
    a. Age:
    b. Gender: [] Male [] Female [] Other
    c. Occupation:

2. Online Payment Usage:
    - How frequently do you make online payments?
        a. Rarely
        b. Occasionally
        c. Frequently

3. Preferred Online Payment Method:
    - Which online payment method do you prefer?
        a. Credit/Debit Card
        b. Mobile Wallet
        c. Net Banking
        d. UPI

4. Security Concerns:
    - How confident are you in the security of online payment platforms?
        a. Very Confident
        b. Somewhat Confident
        c. Neutral
        d. Somewhat Concerned
        e. Very Concerned

5. User-Friendly Interface:
   - Rate the user-friendliness of the online payment platforms you've used (on a scale of 1 to 5, where 1 is not user-friendly, and 5 is very user-friendly).

6. Features Importance:
   - Rank the importance of the following features in an online payment platform (drag and drop to prioritize):
     a. Speed
     b. Security
     c. Ease of Use
     d. Rewards/Incentives

7. Feedback and Improvement:
   - How likely are you to recommend your primary online payment platform to others?
     a. Very Likely
     b. Likely
     c. Neutral
     d. Unlikely
     e. Very Unlikely

Below is the sample data provided for analysis.

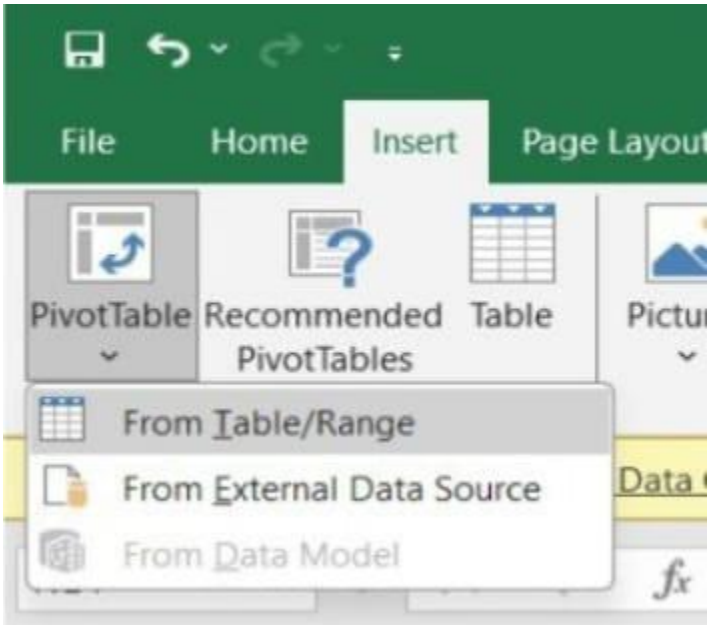
	A	B	C	D	E	F	G	H	I
1	Age	Name	Occupation	Usage	Method used	Security Concerns	User-Friendly	Features	Feedback
2	31	Mohit Singh	Developer	Frequently	Credit/Debit cards	Very Confident	5	Speed, Security, Ease of Use, Rewards/Incentives	Very Likely
3	54	Radhika More	Doctor	Frequently	Net Banking	Very Concerned	4	Security, Ease of Use, Speed, Rewards/Incentives	Likely
4	21	Yash Patil	Student	Occasionally	Mobile Wallet	Somewhat Confident	3	Ease of Use, Speed, Security, Rewards/Incentives	Neutral
5	40	Vikas Rane	Engineer	Frequently	UPI	Somewhat Concerned	4	Speed, Security, Ease of Use, Rewards/Incentives	Likely
6	46	Aarav Patel	Lawyer	Occasionally	Credit/Debit cards	Neutral	3	Security, Ease of Use, Rewards/Incentives, Speed	Unlikely
7	35	Riya Sharma	Sales	Occasionally	Mobile Wallet	Somewhat Concerned	4	Rewards/Incentives, Ease of Use, Speed, Security	Likely
8	30	Advait Singh	Developer	Rarely	Credit/Debit cards	Very Confident	5	Speed, Ease of Use, Rewards/Incentives, Security	Very Likely
9	30	Ishita Verma	Teacher	Frequently	Net Banking	Somewhat Confident	4	Ease of Use, Security, Speed, Rewards/Incentives	Likely
10	48	Aryan Gupta	Engineer	Occasionally	UPI	Very Concerned	3	Security, Speed, Rewards/Incentives, Ease of Use	Unlikely
11	47	Ananya Reddy	Lawyer	Frequently	Mobile Wallet	Neutral	4	Security, Rewards/Incentives, Ease of Use, Speed	Very Likely
12	48	Kabir Desai	Developer	Frequently	Credit/Debit cards	Very Concerned	3	Security, Speed, Ease of Use, Rewards/Incentives	Unlikely
13	39	Meera Nair	Sales	Occasionally	Mobile Wallet	Somewhat Concerned	4	Rewards/Incentives, Ease of Use, Speed, Security	Likely
14	40	Arjun Saha	Engineer	Frequently	UPI	Somewhat Confident	4	Speed, Security, Ease of Use, Rewards/Incentives	Likely
15	18	Diya Choudhury	Student	Rarely	Net Banking	Neutral	2	Ease of Use, Speed, Security, Rewards/Incentives	Very Unlikely
16	28	Rohan Kapoor	Developer	Frequently	Credit/Debit cards	Very Confident	5	Speed, Ease of Use, Rewards/Incentives, Security	Very Likely
17	50	Saanvi Joshi	Teacher	Frequently	Mobile Wallet	Very Concerned	3	Security, Speed, Rewards/Incentives, Ease of Use	Unlikely
18	50	Vihaan Mishra	Engineer	Frequently	UPI	Somewhat Confident	4	Speed, Security, Ease of Use, Rewards/Incentives	Likely
19	49	Anika Banerjee	Doctor	Occasionally	Net Banking	Somewhat Concerned	4	Security, Ease of Use, Speed, Rewards/Incentives	Likely
20	29	Zayn Malhotra	Sales	Occasionally	Mobile Wallet	Very Concerned	3	Rewards/Incentives, Ease of Use, Speed, Security	Unlikely
21	46	Myra Khanna	Lawyer	Frequently	UPI	Neutral	4	Security, Rewards/Incentives, Ease of Use, Speed	Very Likely
22	38	Virat Kumar	Engineer	Frequently	Credit/Debit cards	Neutral	3	Speed, Ease of Use, Rewards/Incentives, Security	Likely
23	40	Aisha Mehra	Teacher	Frequently	Net Banking	Somewhat Confident	4	Ease of Use, Security, Speed, Rewards/Incentives	Likely
24	34	Arnav Bhatia	Developer	Rarely	UPI	Very Confident	2	Security, Speed, Ease of Use, Rewards/Incentives	Very Unlikely
25	47	Ishani Singhania	Doctor	Occasionally	Mobile Wallet	Very Concerned	3	Rewards/Incentives, Ease of Use, Speed, Security	Unlikely
26	25	Krish Patel	Student	Frequently	Credit/Debit cards	Very Confident	4	Speed, Ease of Use, Rewards/Incentives, Security	Likely

Now we have performed 3analysis from the sample data

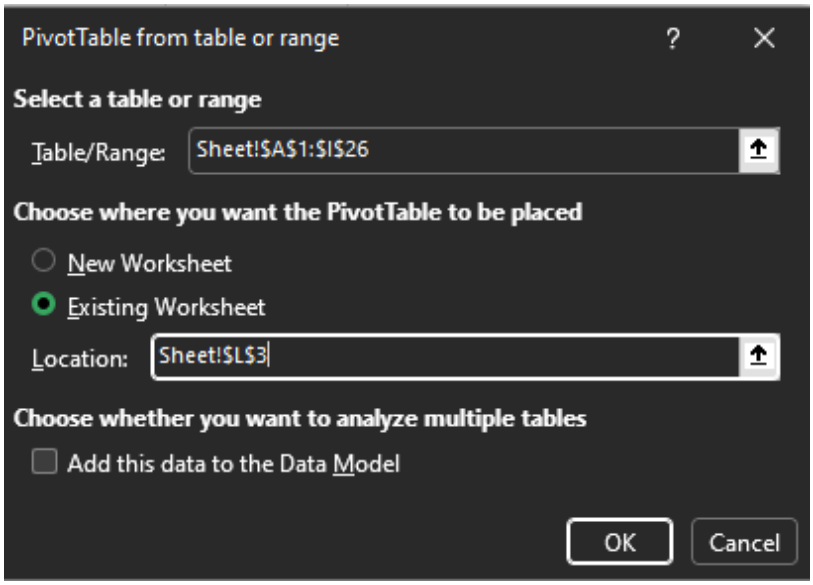
1. **Frequency Distribution Analysis (via Pivot Table):**

The primary goal was to understand the demographic distribution of users, particularly in terms of occupation and their corresponding levels of user-friendliness and security concerns. This analysis provides valuable insights into whether the app is reaching its intended audience. Additionally, it offers suggestions on how the company can alleviate security concerns through the implementation of various methodologies.

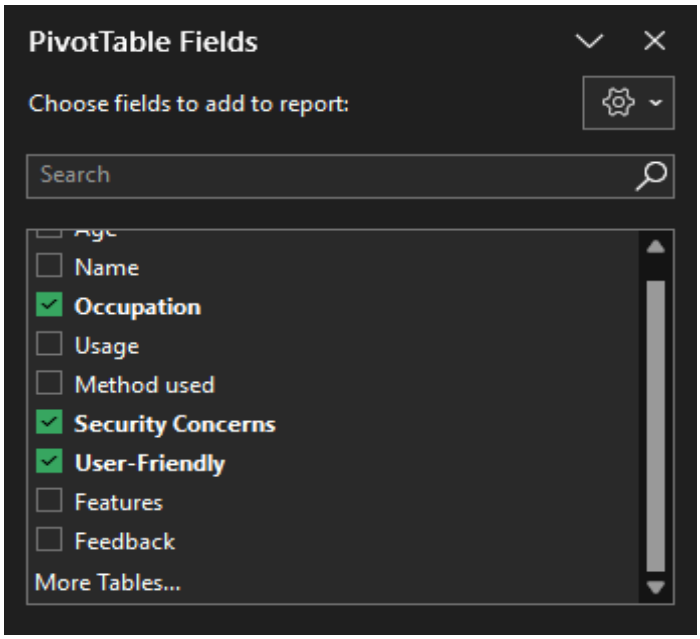
Step 1:- Go to Insert -> PivotTable -> From Table/Range



Step 2:- Select full table, and provide where you want to see result, then click on Ok.



Step 3:- At right-side PivotTable fields will appear, select Occupation, Security Concerns, & User-Friendly.



Output:-

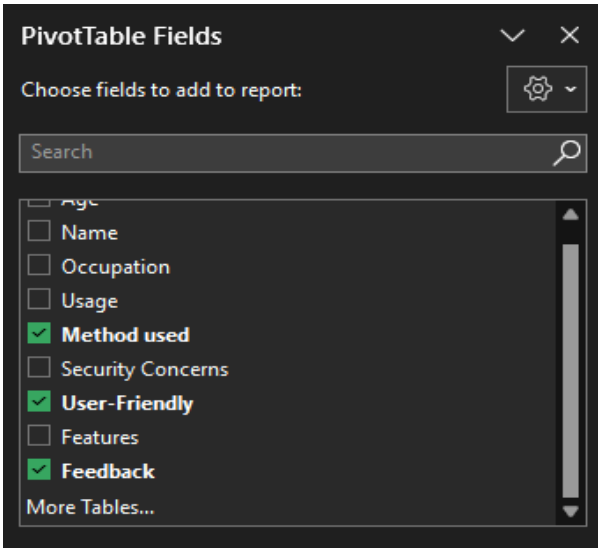
Row Labels	Sum of User-Friendly
Developer	**20**
Very Concerned	3
Very Confident	17
Doctor	**11**
Somewhat Concerned	4
Very Concerned	7
Engineer	**18**
Neutral	3
Somewhat Concerned	4
Somewhat Confident	8
Very Concerned	3
Lawyer	**11**
Neutral	11
Sales	**11**
Somewhat Concerned	8
Very Concerned	3
Student	**9**
Neutral	2
Somewhat Confident	3
Very Confident	4
Teacher	**11**
Somewhat Confident	8
Very Concerned	3
Grand Total	**91**

2. Pivot Table Analysis:

The focus of this analysis was to determine the popularity and user-friendliness of different online payment methods. By identifying the most and least used methods, the company gains insights into user preferences and areas where users may find certain methods challenging. This knowledge is instrumental in streamlining the payment process and improving user satisfaction.

Step 1& 2: Both are same as frequency distribution.

Step 3: At right-side PivotTable Fields will appear, select Method used, Feedback & User-Friendly.



Output:-

Row Labels	Sum of User-Friendly
Credit/Debit cards	**28**
Likely	7
Unlikely	6
Very Likely	15
Mobile Wallet	**24**
Likely	8
Neutral	3
Unlikely	9
Very Likely	4
Net Banking	**18**
Likely	16
Very Unlikely	2
UPI	**21**
Likely	12
Unlikely	3
Very Likely	4
Very Unlikely	2
Grand Total	**91**

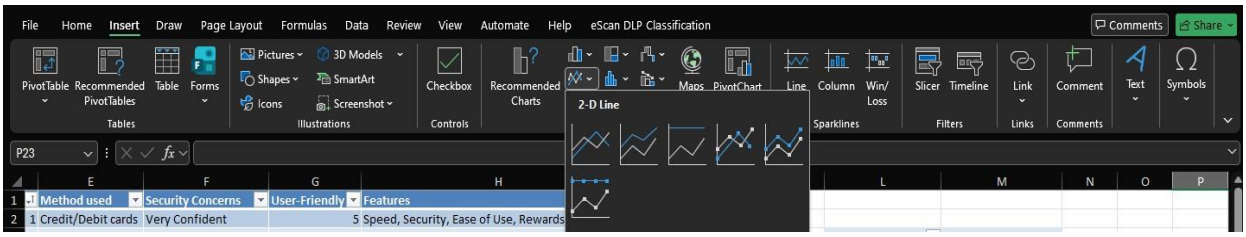
3. Scatter Plot Analysis:

The scatter plot was generated to explore the relationship between user-friendliness ratings and usage patterns. This visualization aids in understanding how an increase in user-friendliness can potentially enhance usage and attract more customers. The plotted data points provide a clear picture of the correlation between these two essential factors, guiding the company in stratergies to expand its user base.

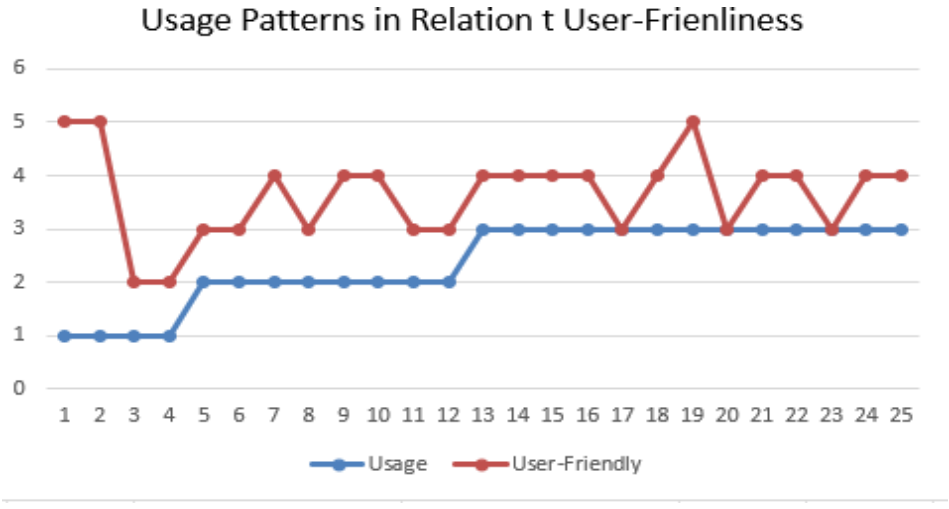
Step 1:- Convert usage to rating (by assuming Rarely=1, Occasionally=2 & Frequently=3) and put it in a separate column, we will get something like this.

Usage
1
1
1
1
2
2
2
2
2
2
2

Step 2:- Now select Usage rating & User-Friendly columns, then go to Insert -> Click on Line or Area chart options -> Then select “Line with Markers” option.



Output:-



Practical 3(b)

Aim:- Perform analysis of given secondary data.

1. **Determine your research question-** Knowing exactly what you are looking for.

2. **Locating data-** Knowing what is out there and whether you can gain access to it. A quick Internet search, possibly with the help of a librarian, will reveal a wealth of options.

3. **Evaluating relevance of the data -** Considering things like the data's original purpose, when it was collected, population, sampling strategy/sample, data collection protocols, operationalization of concepts, questions asked, and form/shape of the data.

4. **Assessing credibility of the data -** Establishing the credentials of the original researchers, searching for full explication of methods including any problems encountered, determining how consistent the data is with data from other sources, and discovering whether the data has been used in any credible published research.

5. **Analysis -** This will generally involve a range of statistical processes.

Example: Analyze the given Population Census Data for Planning and Decision Making by using the size and composition of populations.

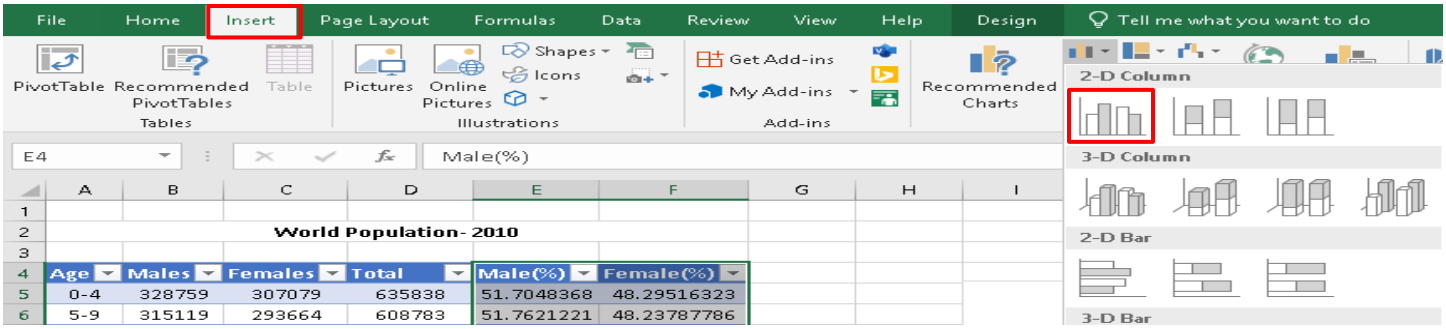
	A	B	C	D	E	F
1						
2	World Population- 2010					
3						
4	Age	Males	Females	Total	Male(%)	Female(%)
5	0-4	328759	307079	635838		
6	5-9	315119	293664	608783		
7	10-14	311456	290598	602054		
8	15-19	312831	293313	606144		
9	20-24	311077	295739	606816		
10	25-29	284258	273379	557637		
11	30-34	255596	247383	502979		
12	35-39	248575	241938	490513		
13	40-44	232217	226914	459131		
14	45-49	202633	201142	403775		
15	50-54	176241	176440	352681		
16	55-59	153494	156283	309777		
17	60-64	114194	121200	235394		
18	65-69	83129	92071	175200		
19	70-74	65266	77990	143256		
20	75-79	43761	56895	100656		
21	80-84	25060	37873	62933		
22	85+	14164	28156	42320		

Put the cursor in cell **D22** and click on the **AutoSum** and then click **Enter**. This will calculate the total population. Then copy the formula in cell **E22** across the row **22**. To calculate the percent of males in cell **E5**, enter the formula =**1*100*B5/$D$22**. And copy the formula in cell **E5** down to cell **E22**.

To calculate the percent of females in cell **F4**, enter the formula **=100*C5/SD$22**. Copy the formula in cell **F5** down to cell **F22**

	A	B	C	D	E	F
1						
2	World Population- 2010					
3						
4	Age	Males	Females	Total	Male(%)	Female(%)
5	0-4	328759	307079	635838	51.7048368	48.29516323
6	5-9	315119	293664	608783	51.7621221	48.23787786
7	10-14	311456	290598	602054	51.7322366	48.26776336
8	15-19	312831	293313	606144	51.6100135	48.38998654
9	20-24	311077	295739	606816	51.2638098	48.73619021
10	25-29	284258	273379	557637	50.9754554	49.02454464
11	30-34	255596	247383	502979	50.8164357	49.18356432
12	35-39	248575	241938	490513	50.6765366	49.32346339
13	40-44	232217	226914	459131	50.577504	49.42249598
14	45-49	202633	201142	403775	50.1846325	49.81536747
15	50-54	176241	176440	352681	49.9717875	50.02821246
16	55-59	153494	156283	309777	49.5498375	50.45016254
17	60-64	114194	121200	235394	48.5118567	51.48814328
18	65-69	83129	92071	175200	47.4480594	52.55194064
19	70-74	65266	77990	143256	45.5589993	54.44100073
20	75-79	43761	56895	100656	43.4757988	56.52420124
21	80-84	25060	37873	62933	39.8201262	60.17987383
22	85+	14164	28156	42320	33.4688091	66.53119093
23	**Total**	**3477830**	**3418057**	**6895887**	**50.433396**	**49.56660398**

To build the population pyramid , we need to choose a horizontal bar chart with two series of data (% male and % female) and the age labels in column A as the **Category X-axis** labels. Highlight the range **A5:A22**, hold down the **CTRL** key and highlight the range **E5:F22.** Under **Insert** tab, under horizontal bar charts select **clustered bar chart.**

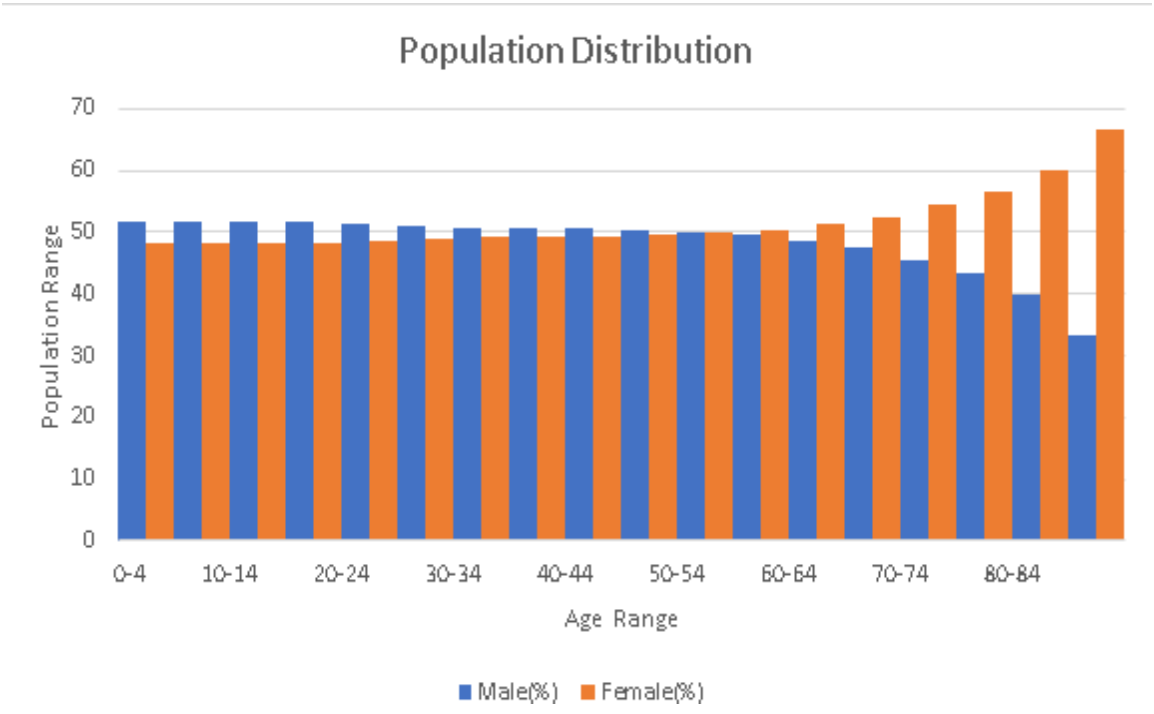


Put the tip of your mouse arrow on the **Y-axis** (vertical axis) so it says – Category Axis, right click and choose **Format Axis**.

Choose **Axis Options** tab and set the major and minor tick mark type to None, Axis labels to **Low** and Click **Ok**.

Click on any of the bars in your pyramid, click right and select – format data series. Set the **Overlap** to **100** and **Gap Width** to **0**. Click **Ok**.

Output:-



Practical 4(a)

Aim:- Perform Testing of Hypothesis using two sample t-test.

The T distribution provides a good way to perform one sample tests on the mean when the population variance is not known provided the population is normal or the sample is sufficiently large so that the Central Limit Theorem applies.

Two Sample t-test

Example:- A college principal informed classroom teachers that some of their students showed unusual potential for intellectual gains. One month later the students identified to teachers as having potential for unusual intellectual gains showed significantly greater gains performance on a test set to measure IQ than did students who were not so identified. Below are the data for students:

Experimental	Comparison	Column1
35	2	
40	27	
12	38	
15	31	
21	1	
14	19	
46	1	
10	34	
28	3	
48	1	
16	2	
30	3	
32	2	
48	1	
31	2	
22	1	
12	3	
39	29	
19	37	
25	2	
27.15	11.95	**Mean**
12.50799744	14.61244963	**Std Dev**

- **Experimental Data**

To calculate Standard Mean go to cell A22 and type `**=SUM(A2:A21)/20**

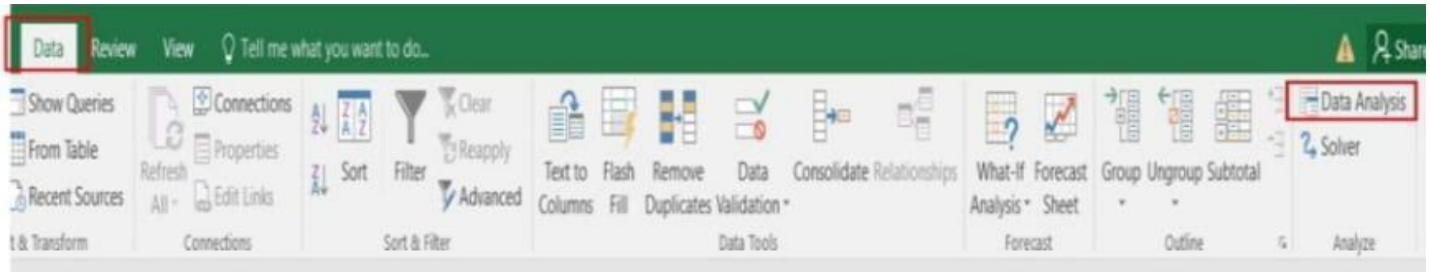
To calculate Standard Deviation go to cell A23 and type **=STDEV(A2:A21)**

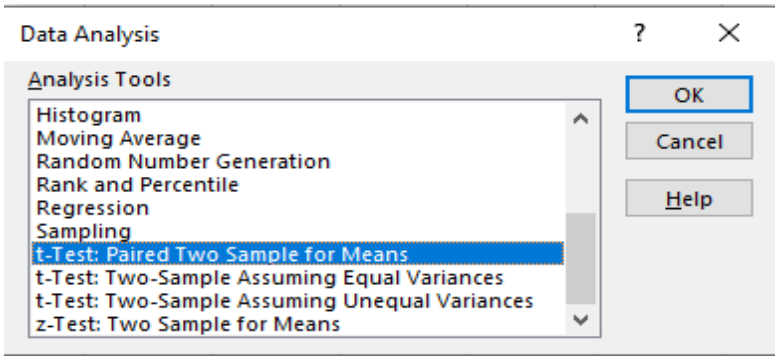
- **Comparison Data**

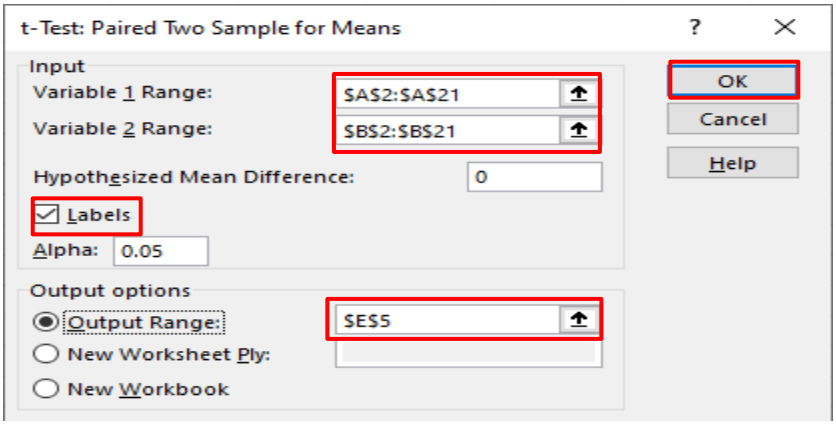
To calculate Standard Mean go to cell A22 and type `**=SUM(B2:B21)/20**

To calculate Standard Deviation go to cell A23 and type **=STDEV(B2:B21)**

To find T-Test Statistics by going through Data then selecting Data Analysis







To calculate the T-Test square value go to cell F20 and type

=(A22-B22)/SQRT((A23*A23)/COUNT(A2:A21)+(B23*B23)/COUNT(A2:A21))

Now go to cell E22 and type

=IF(E20<E12, "H0 is Accepted", "H0 is Rejected and H1 is Accepted")

Our calculated value is larger than the tabled value at alpha = 01, so we reject the null hypothesis and accept the alternative hypothesis, namely, that the difference in gain scores is likely the result of the experimental treatment and not the result of chance variation.

Output:-

	A	B	C	D	E	F	G
1	Experimental	Comparison	Column1				
2	35	2					
3	40	27					
4	12	38					
5	15	31			t-Test: Paired Two Sample for Means		
6	21	1					
7	14	19				*35*	*2*
8	46	1			Mean	26.73684211	12.47368421
9	10	34			Variance	161.5380117	219.5964912
10	28	3			Observations	19	19
11	48	1			Pearson Correlation	-0.38128717	
12	16	2			Hypothesized Mean Difference	0	
13	30	3			df	18	
14	32	2			t Stat	2.714013677	
15	48	1			P(T<=t) one-tail	0.007110878	
16	31	2			t Critical one-tail	1.734063607	
17	22	1			P(T<=t) two-tail	0.014221756	
18	12	3			t Critical two-tail	2.10092204	
19	39	29					
20	19	37			Calculated Value:	3.534053898	
21	25	2					
22	27.15	11.95	**Mean**		H0 is Rejected and H1 is Accepted		
23	12.50799744	14.61244963	**Std Dev**				
24							

Practical 4(b)

Aim:- Perform Testing of Hypothesis using paired t-test.

The T distribution provides a good way to perform one sample tests on the mean when the population variance is not known provided the population is normal or the sample is sufficiently large so that the Central Limit Theorem applies.

Paired t-test

Example: A college principal informed classroom teachers that some of their students showed unusual potential for intellectual gains. One month later the students identified to teachers as having potential for unusual intellectual gains showed significantly greater gains per performance on a test said to measure IQ than did students who were not so identified. Below are the data for the students:

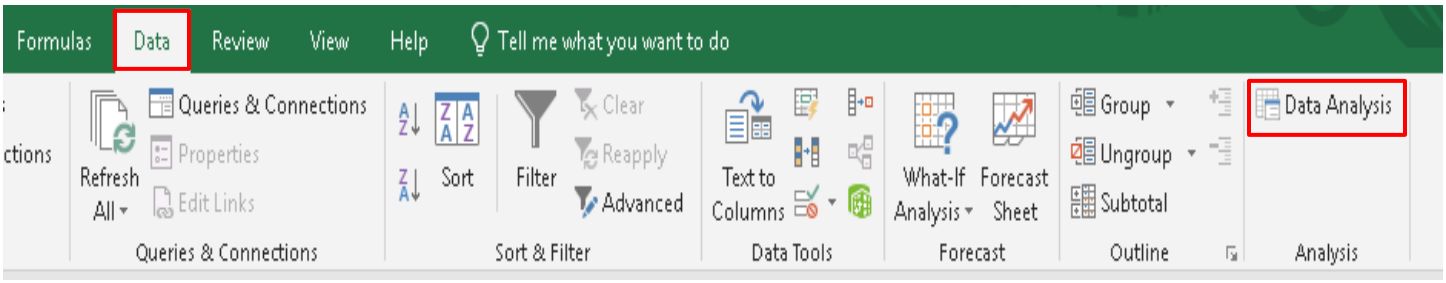
Experimental	Comparison		Difference	
35	2		33	
40	27		13	
12	38		-26	
15	31		-16	
21	1		20	
14	19		-5	
46	1		45	
10	34		-24	
28	3		25	
48	1		47	
16	2		14	
30	3		27	
32	2		30	
48	1		47	
31	2		29	
22	1		21	
12	3		9	
39	29		10	
19	37		-18	
25	2		23	
27.15	11.95	Mean	15.2	Mean of Differences
12.50799744	14.61244963	Std Dev	22.686885	Std Dev of Differences

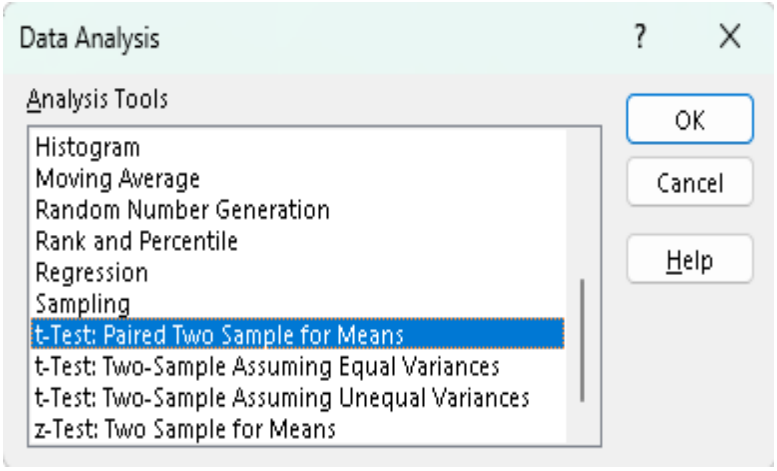
- **Difference Data**

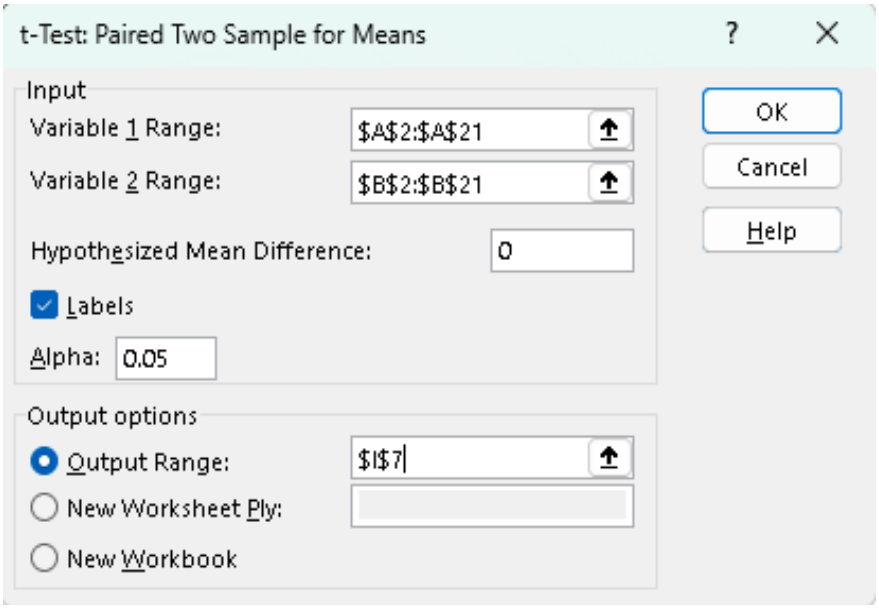
To calculate the **Mean of Differences,** go to cell C22 and type = AVERAGE(C2:C21)

To calculate the **Standard Deviation of Differences**, go to cell C23 and type = STDEV(C2:C21)

To perform the Paired t-test, go to **Data -> Data Analysis -> t-Test: Paired Two Sample for Means.**







To calculate the Paired t-test Statistic, go to cell E20 and type

=AVERAGE(C2:C21)/(STDEV(C2:C21)/SQRT(COUNT(C2:C21)))

Now go to cell F22 and type

=IF(F20<F18, "H0 is Accepted", "H0 is Rejected and H1 is Accepted")

Since: Calculated t-value = **2.996289153** > Critical t-value (two-tail) **= 2.860934606**

Therefore: H0 is Rejected and H1 is Accepted.

Conclusion: There is a Statistically Significant difference between the Experimental and Comparison scores at the **1% level of significance**. Hence, the difference in the scores is significant and is unlikely to have occurred due to chance variation.

Experimental	Comparison		Difference	
35	2		33	
40	27		13	
12	38		-26	
15	31		-16	
21	1		20	
14	19		-5	
46	1		45	
10	34		-24	
28	3		25	
48	1		47	
16	2		14	
30	3		27	
32	2		30	
48	1		47	
31	2		29	
22	1		21	
12	3		9	
39	29		10	
19	37		-18	
25	2		23	
27.15	11.95	Mean	15.2	Mean of Differences
12.50799744	14.61244963	Std Dev	22.686885	Std Dev of Differences
			2.9962892	Paired t-test statistics

t-Test: Paired Two Sample for Means

	Variable 1	*Variable 2*
Mean	26.73684211	12.47368421
Variance	161.5380117	219.5964912
Observations	19	19
Pearson Correlation	-0.38128717	
Hypothesized Mean Difference	0	
df	18	
t Stat	2.714013677	
P(T<=t) one-tail	0.007110878	
t Critical one-tail	1.734063607	
P(T<=t) two-tail	0.014221756	
t Critical two-tail	2.10092204	
Calculated Value:	2.996289153	
H0 is Rejected and H1 is Accepted		

Practical 5(a)

Aim:- Perform testing of hypothesis using Chi-Squared Goodness-of-Fit test.

A system administrator need to upgrade the computers for his division. HE wants to know what sort of computer system his workers prefer. He gives three choices Windows, MSc, Linux. Test the hypothesis or theory that an equal percentage of the population prefers each type of computer systems.

System	O	Ei
Windows	20	33.33
Mac	60	33.33
Linux	20	33.33
Total	100	99.99

Example: From Total Students, 100 student marks are given in the table below, create a periodic sample consisting of every 10th student.

You need to run the sampling data analysis tool to create a sample by selecting the 100 cells in the marks column as Input range and Periodic with Period 10.

H0: The population distribution of the variable is as same as the proposed distribution.

H1: The distributions are different.

To calculate the **Chi-Squared value for Windows** go to cell D2 and type

=(B2-C2)*(B2-C2))/C2

To calculate the **Chi-Squared value for Mac** go to cell D2 and type

=(B3-C3)*(B3-C3))/C3

To calculate the **Chi-Squared value for Linux** go to cell D2 and type

=(B4-C4)*(B4-C4))/C4

Now,

Go to cell D5 for $\sum \frac{Oi - Ei}{E_i}$ and type =SUM(D2:D4)

To get the table value for Chi-Square for α = 0.05 and dof = 2, go to cell A9 and type = CHIINV(0.95,2)

At cell D8 type = IF(D5>G5, "H0 is accepted and H1 is Rejected" , "H0 is Rejected and H1 is Accepted")

Output:

System	O	Ei	Chi-Squared value
Windows	20	33.33	5.33120012
Mac	60	33.33	21.34080108
Linux	20	33.33	5.33120012
Total	100	99.99	32.00320132

H0 = The distribution of variables is same as proposed distribution.

H1 = the distribution of the variables is not same as proposed distribution.

Total Value: 5.991465

H0 is Accepted & H1 is Rejected

Practical 5(b)

<u>Aim</u>:- Perform testing of hypothesis using Chi-Squared test of Independence.

In a study to understand the performance of MSc Ds Part – I class, a college selects a random sample of 100 students. Each student was asked his grade obtained in BSc DS. The sample is as given below

Roll No	Student's Name	Gender	Grade	Roll No	Student's Name	Gender	Grade	Roll No	Student's Name	Gender	Grade
1	Aanya Rao	Female	A	30	Anaya Pillai	Female	A	73	Raj Menon	Male	O
2	Diya Desai	Female	B	31	Bhavya Malhotra	Female	A	74	Veer Mukherjee	Male	B
3	Isha Malhotra	Female	B	32	Chhavi Ghosh	Female	C	75	Aarush Iyer	Male	C
4	Kavya Reddy	Female	O	33	Devika Pandey	Female	C	76	Atharv Mehta	Male	O
5	Myra Singh	Female	O	34	Eva Mehta	Female	C	77	Dev Yadav	Male	A
6	Riya Mishra	Female	A	35	Fiza Chauhan	Female	B	78	Harsh Mukherjee	Male	C
7	Saanvi Verma	Female	A	36	Gia Desai	Female	A	79	Kartik Sinha	Male	A
8	Tara Chopra	Female	B	37	Ishita Mukherjee	Female	B	80	Manav Rao	Male	C
9	Zara Tiwari	Female	O	38	Kyra Bose	Female	B	81	Nikhil Menon	Male	B
10	Anika Verma	Female	O	39	Larisa Mehta	Female	D	82	Pranav Patel	Male	B
11	Avni Joshi	Female	O	40	Mahi Ghosh	Female	C	83	Ritvik Dutta	Male	A
12	Charvi Khanna	Female	B	41	Nitya Chatterjee	Female	B	84	Samar Singh	Male	D
13	Disha Naidu	Female	O	42	Ojal Agarwal	Female	A	85	Tanish Kumar	Male	O
14	Esha Desai	Female	A	43	Pihu Malhotra	Female	A	86	Uday Nair	Male	B
15	Gauri Mehta	Female	D	44	Ridhi Agarwal	Female	D	87	Varun Rastogi	Male	O
16	Hiya Bansal	Female	C	45	Simran Desai	Female	A	88	Yuvraj Mehta	Male	C
17	Ira Iyer	Female	B	46	Trisha Chopra	Female	O	89	Zain Yadav	Male	O
18	Jiya Bansal	Female	A	47	Vidya Agarwal	Female	B	90	Aakash Mishra	Male	B
19	Kiara Singh	Female	D	48	Yamini Sinha	Female	D	91	Bhargav Bose	Male	B
20	Lavanya Reddy	Female	C	49	Aditi Pandey	Female	B	92	Chirag Menon	Male	D
21	Meera Chauhan	Female	O	50	Bela Bhatt	Female	B	93	Deepak Dubey	Male	D
22	Navya Bhatt	Female	A	51	Aarav Kapoor	Male	O	94	Eshan Yadav	Male	D
23	Prisha Pandey	Female	O	52	Vivaan Bose	Male	C	95	Farhan Rao	Male	B
24	Rhea Rao	Female	D	53	Aditya Kulkarni	Male	D	96	Girish Malhotra	Male	A
25	Sana Singh	Female	O	54	Vihaan Kulkarni	Male	D	97	Hemant Bansal	Male	A
26	Tanvi Pillai	Female	O	55	Arjun Dubey	Male	B	98	Ivan Kulkarni	Male	O
27	Urvi Dubey	Female	O	56	Sai Nair	Male	O	99	Jatin Malhotra	Male	O
28	Vanya Joshi	Female	A	57	Reyansh Sinha	Male	A	100	Kunal Malhotra	Male	A
29	Yashvi Gupta	Female	B	58	Ayaan Patel	Male	A				
30	Anaya Pillai	Female	A								

Null Hypothesis H0: The performance of Girls students is same as Boys students.

Alternate Hypothesis H1: The performance of Girls and Boys students are different.

Open Excel Workbook:-

	O	A	B	C	D	Total	$\sum (O_i - E_i)^2 / E_i$
Girls	12	13	13	6	6	50	
Boys	13	11	10	7	9	50	
Total	25	24	23	13	15	100	
∑i							

Prepare a contingency table as shown above

- To calculate Girls students with “O” Grade:

Go to Cell H7 and type =COUNTIF(D2:D51,"O")

- To calculate Boys students with “O” Grade:

Go to cell H8 and type =COUNTIF(D52:D101,"O")

- To calculate Girls students with “A” Grade:

Go to cell I7 and type =COUNTIF(D2:D51,"A")

- To calculate Boys students with “A” Grade:

Go to cell I8 and type =COUNTIF(D52:D101,"A")

- To calculate Girls students with “B” Grade:

Go to cell J7 and type =COUNTIF(D2:D51,"B")

- To calculate Boys students with “B” Grade:

Go to cell J8 and type =COUNTIF(D52:D101,"B")

- To calculate Girls students with “C” Grade:

Go to cell K7 and type =COUNTIF(D2:D51,"C")

- To calculate Boys students with “C” Grade:

Go to cell K8 and type =COUNTIF(D52:D101,"C")

- To calculate Girls students with “D” Grade:

Go to cell L7 and type =COUNTIF(D2:D51,"D")

- To calculate Boys students with “A” Grade:

Go to cell L8 and type =COUNTIF(D52:D101,"D")

To calculate the Expected Value Ei

Go to cell H10 and type =50*25/100

Go to cell I10 and type =50*24/100

Go to cell J10 and type =50*23/100

Go to cell K10 and type =50*13/100

Go to cell L10 and type =50*15/100

And calculate,

Total Expected Value =SUM(H10,I10,J10,K10,L10)

To calculate $\sum (O_i - E_i)^2 / E_i$

Go to cell N7 and type =SUM((I7-I$10)^2/I$10)+((J7-J$10)^2/J$10)+((K7-K$10)^2/K$10)+((L7-L$10)^2/L$10)+((M7-M$10)^2/M$10)

Go to cell N8 and type =SUM((I8-I$10)^2/I$10)+((J8-J$10)^2/J$10)+((K8-K$10)^2/K$10)+((L8-L$10)^2/L$10)+((M8-M$10)^2/M$10)

For total go to cell N9 and type =N7+N8

To find Critical Value go to cell M13 and type =CHISQ.INV.RT(0.05, 4)

To fond the Final Decision go to cell M15 and type =IF(N9 > N11, "H0 is Accepted", "H0 is Rejected")

Output:

	O	A	B	C	D	Total	$\sum (O_i - E_i)^2 / E_i$
Girls	12	13	13	6	6	50	0.617447046
Boys	13	11	10	7	9	50	0.617447046
Total	25	24	23	13	15	100	1.234894091
∑i	12.5	12	11.5	6.5	7.5	50	
					Critical Value	9.487729037	
					Decision	H0 is Accepted	

Practical 6

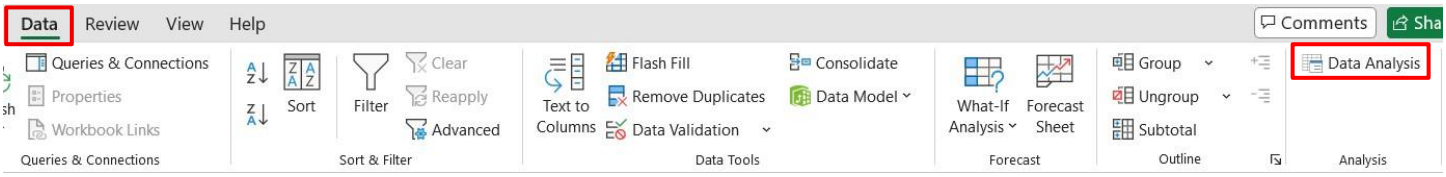
<u>Aim</u>:- Perform testing of hypothesis using Z – test.

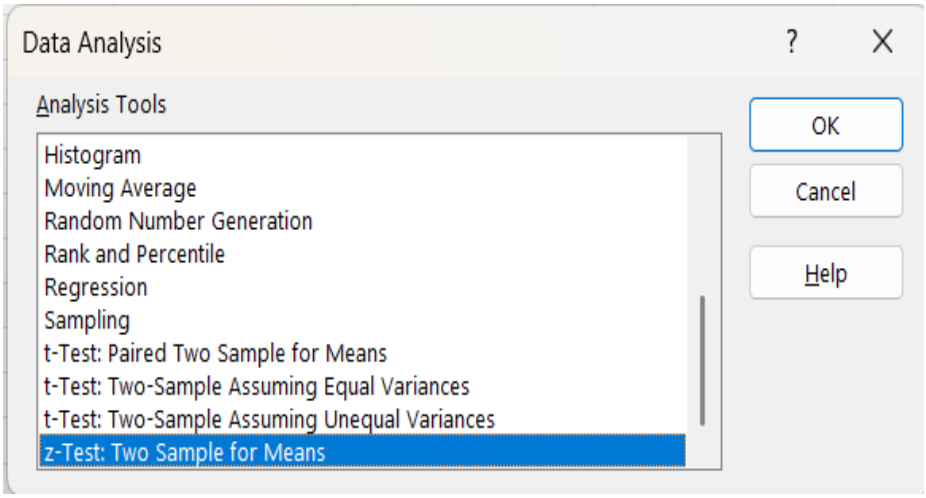
Use a Z - test if:

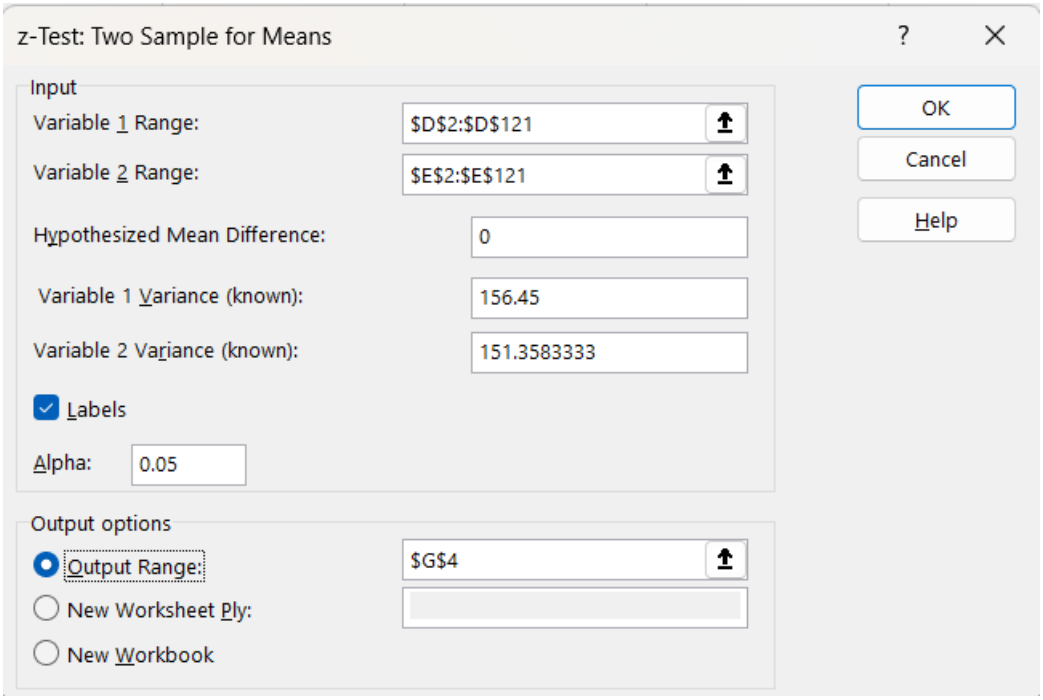
- Your sample size is greater than 30. Otherwise, use a t test.
- Data points should be independent from each other. In other words, one data point isn't related or doesn't affect another data point.
- Your data should be normally distributed. However, for large sample sizes (over 30) this doesn't always matter.
- Your data should be randomly selected from a population, where each item has an equal chance of being selected.
- Sample sizes should be equal if at all possible.

H0: mean of two group is 0.

H1: mean of two group is not 0.







Now, Go to cell H17 and type: **=IF(ABS(H11)>=H15,"H0 is Rejected","H0 is Accepted")**

Output:

	A	B	C	D	E	F	G	H	I
1	patient	gender	agegrp	bp_before	bp_after				
2	1	Male	30-45	143	153				
3	2	Male	30-45	163	170				
4	3	Male	30-45	153	168		z-Test: Two Sample for Means		
5	4	Male	30-45	153	142				
6	5	Male	30-45	146	141			*bp_before*	*bp_after*
7	6	Male	30-45	150	147		Mean	156.5630252	151.3445378
8	7	Male	30-45	148	133		Known Variance	156.45	151.3583333
9	8	Male	30-45	153	141		Observations	119	119
10	9	Male	30-45	153	131		Hypothesized Mean D	0	
11	10	Male	30-45	158	125		z	3.244725283	
12	11	Male	30-45	149	164		P(Z<=z) one-tail	0.00058782	
13	12	Male	30-45	173	159		z Critical one-tail	1.644853627	
14	13	Male	30-45	165	135		P(Z<=z) two-tail	0.00117564	
15	14	Male	30-45	145	159		z Critical two-tail	1.959963985	
16	15	Male	30-45	143	153				
17	16	Male	30-45	152	126		**Decision:**	H0 is Rejected	
18	17	Male	30-45	141	162				
19	18	Male	30-45	176	134				
20	19	Male	30-45	143	136				
21	20	Male	30-45	162	150				
22	21	Male	46-59	149	168				
23	22	Male	46-59	156	155				
24	23	Male	46-59	151	136				
25	24	Male	46-59	159	132				
26	25	Male	46-59	164	160				

Practical 7(a)

Aim:- Perform testing of hypothesis using One way ANOVA.

ANOVA ASSUMPTIONS

- The dependent variable (SAT scores in our example) should be continuous
- The independent variables (districts in our example) should be two or more categorical groups.
- There must be different participants in each group with no participant being in more than one group. In our case, each school cannot be in more than one district.
- The dependent variable should be approximately normally distributed for each category,
- Variances of each group are approximately equal.

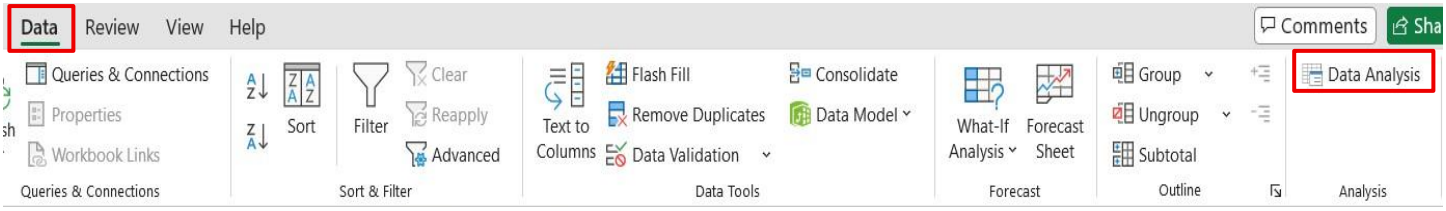
From our data exploration, we can see that the average SAT scores are quite different foreach district. Since we have five different groups, we cannot use the t-test, use the 1-way ANOVA test anyway just to understand the concepts.

H0-There are no significant differences between the Subject's mean SAT scores.

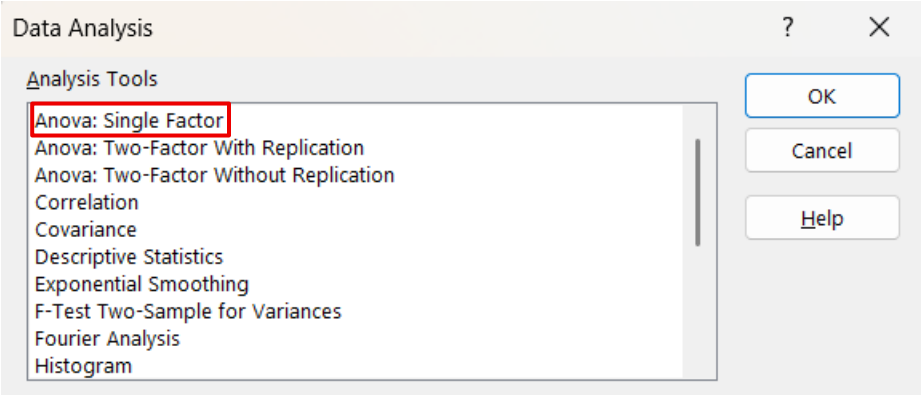
$$\mu 1 = \mu 2 = \mu 3 - 14 - 15$$

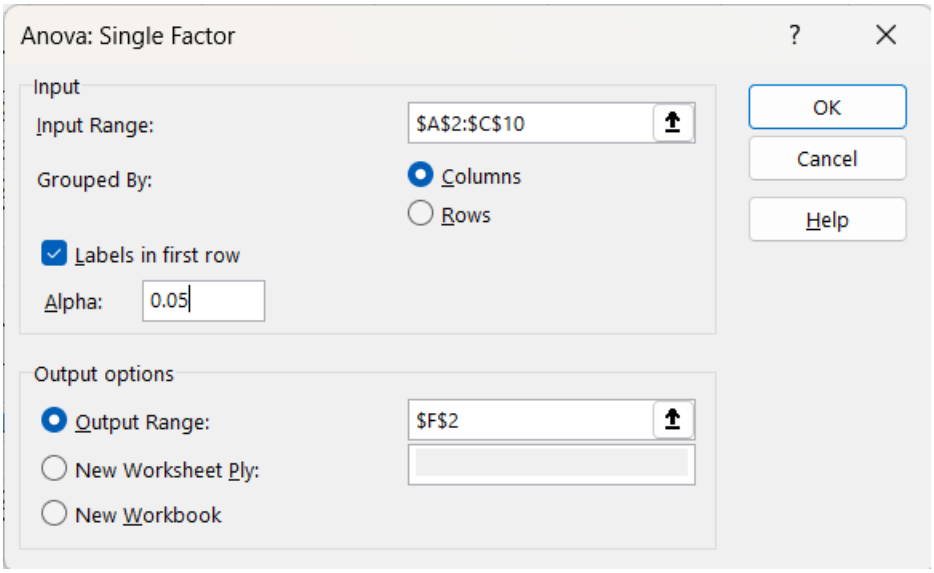
HI-There is a significant difference between the Subject's mean SAT scores.

To perform ANOVA go to data, then select Data Analysis



Select Anova Single Factor -> Then provide input range, select Grouped by Columns, select labels in 1st row & then finally provide output range.





Input Range: $B$1:$D$10 (*Select columns to be analyzed in group*)

Output Range: $F$2 (*Can be any Range*)

Biology	Physics	Chemistry	Column1
45	47	30	
23	16	45	
7	44	35	
49	32	23	
20	26	33	
34		45	
56		33	
23		37	
32.125	33	35.125	Average

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
45	8	244.125	30.515625	254.6269531
47	5	151	30.2	105.2
30	8	286.125	35.765625	50.18945313

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	143.1920015	2	71.59600074	0.504490321	0.612101002	3.554557146
Within Groups	2554.514844	18	141.9174913			
Total	2697.706845	20				

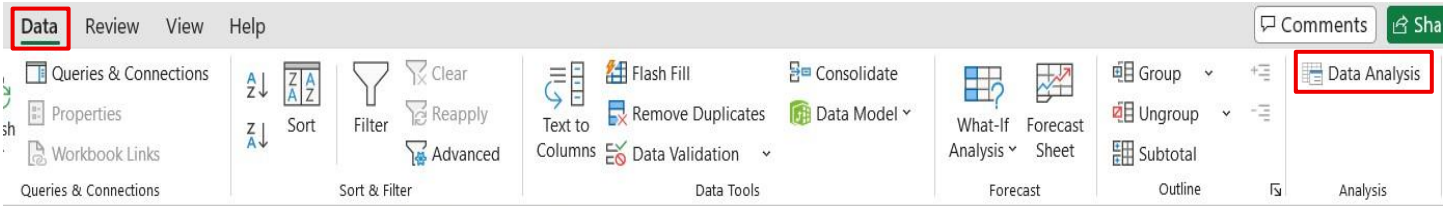
Decision: H0 is Rejected & H1 is Accepted

Since the resulting p-value is less than 0.05. The null hypothesis is rejected and conclude that there is a significant difference between the SAT scores for each subject.

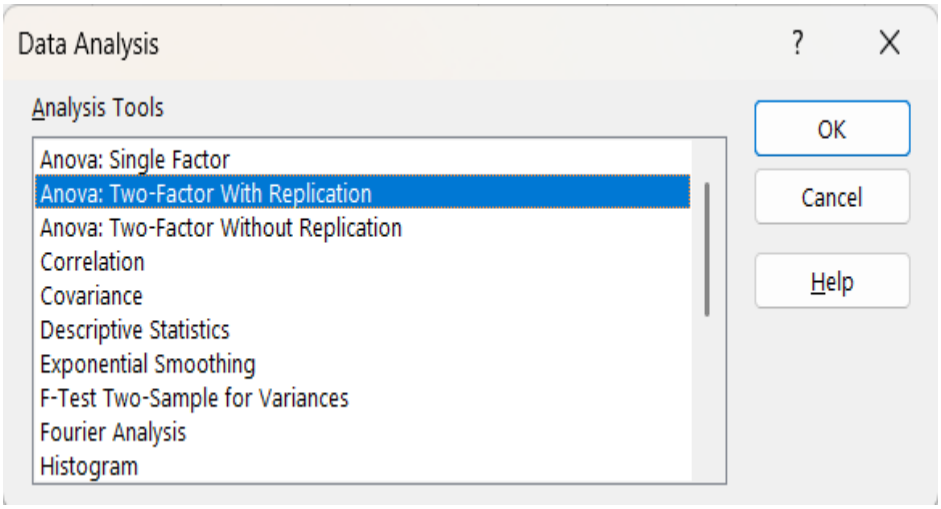
Practical 7(b)

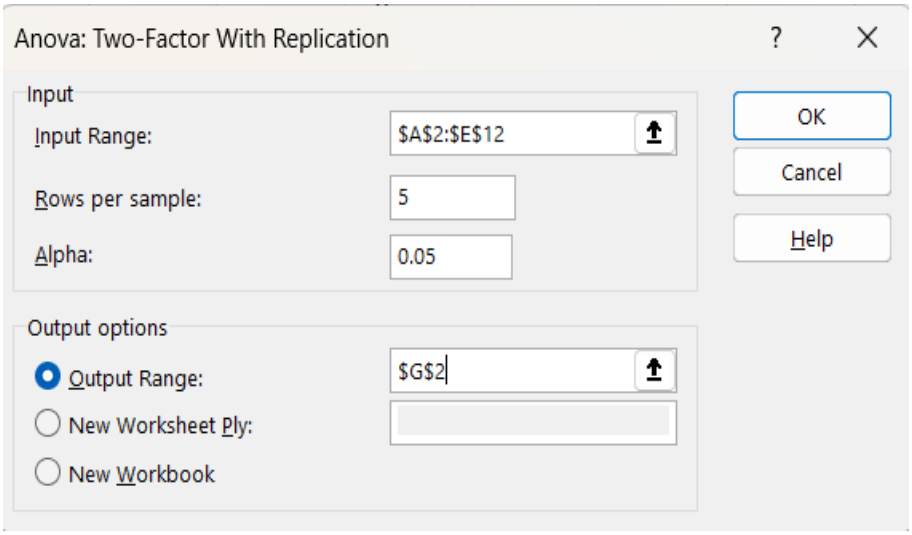
<u>Aim</u>:- Perform testing of hypothesis using Two way ANOVA.

To perform ANOVA, go to Data, then select Data Analysis.



Select Anova Two Factor with Replication -> Then provide input range, rows per column, value of alpha (default is 0.05) and finally provide output range.





Output will look something like below;

Column1	Sunlight Exposure	Column2	Column3	Column4
Watering Frequency	None	Low	Medium	High
Daily	4.8	5	6.4	6.3
	4.4	5.2	6.2	6.4
	3.2	5.6	4.7	5.6
	3.9	4.3	5.5	4.8
	4.4	4.8	5.8	5.8
Weekly	4.4	4.9	5.8	6
	4.2	5.3	6.2	4.9
	3.8	5.7	6.3	4.6
	3.7	5.4	6.5	5.6
	3.9	4.8	5.5	5.5

Decision: H0 is Rejected & H1 is Accepted

Anova: Two-Factor With Replication

SUMMARY	None	Low	Medium	High	Total
Daily					
Count	5	5	5	5	20
Sum	20.7	24.9	28.6	28.9	103.1
Average	4.14	4.98	5.72	5.78	5.155
Variance	0.378	0.232	0.447	0.412	0.775237
Weekly					
Count	5	5	5	5	20
Sum	20	26.1	30.3	26.6	103
Average	4	5.22	6.06	5.32	5.15
Variance	0.085	0.137	0.163	0.317	0.722632
Total					
Count	10	10	10	10	
Sum	40.7	51	58.9	55.5	
Average	4.07	5.1	5.89	5.55	
Variance	0.211222	0.18	0.303222	0.382778	

ANOVA

Source of Varia	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Sample	0.00025	1	0.00025	0.000921	0.975975	4.149097
Columns	18.76475	3	6.254917	23.04898	3.9E-08	2.90112
Interaction	1.01075	3	0.336917	1.241517	0.310898	2.90112
Within	8.684	32	0.271375			
Total	28.45975	39				

Practical 8(a)

Aim:- Perform the Random Sampling and Analyze it.

Example: From total students, 100 student's marks are given in the table below, create a random sample of 10 students.

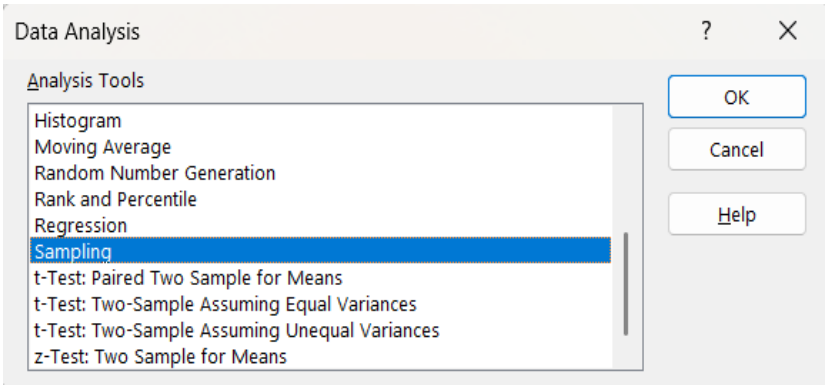
You need to run the sampling Data Analysis Tool to create a random sample of 10 students by selecting all 100 students cells as the Input Range and Random as Sampling Method with 10 for the Random Number of Samples.

Open Existing Excel Sheet with population Data Sample Sheet as given below:

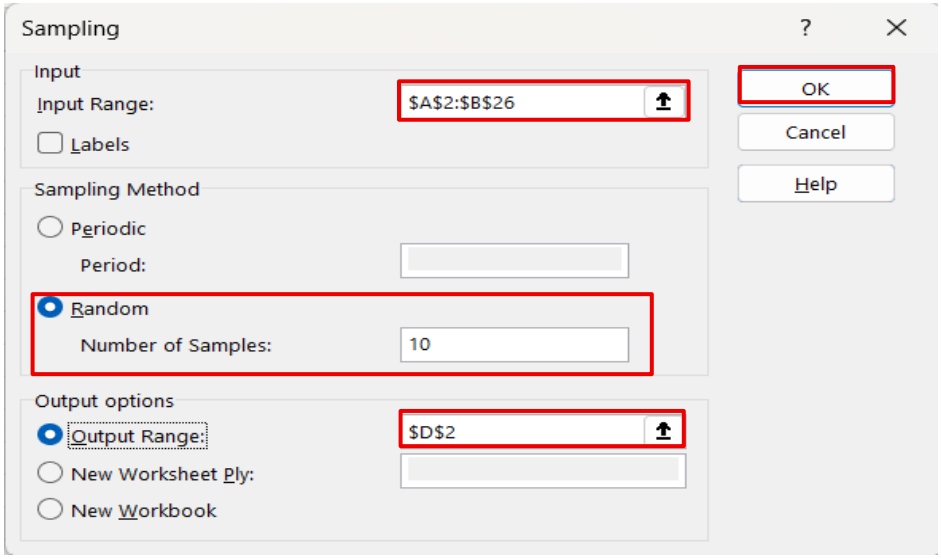
For Random Sampling:

To Perform this,

Go to Data Tab, Click Data Analysis, Scroll and Select Sampling and Click OK.



Then, Select the Input Range, Select Random Radio Button and type 10, then select the Output Range.



Output:

Candidate ID	Test Score		Random 10 Samples for Test Score
1	45		14
2	27		22
3	33		16
4	5		5
5	29		28
6	35		26
7	7		5
8	40		6
9	2		39
10	24		9
11	39		
12	18		
13	14		
14	10		
15	17		
16	13		
17	9		
18	28		
19	32		
20	4		
21	22		
22	2		
23	31		
24	26		
25	24		

Practical 8(b)

Aim:- Perform the Periodic Sampling and Analyze it.

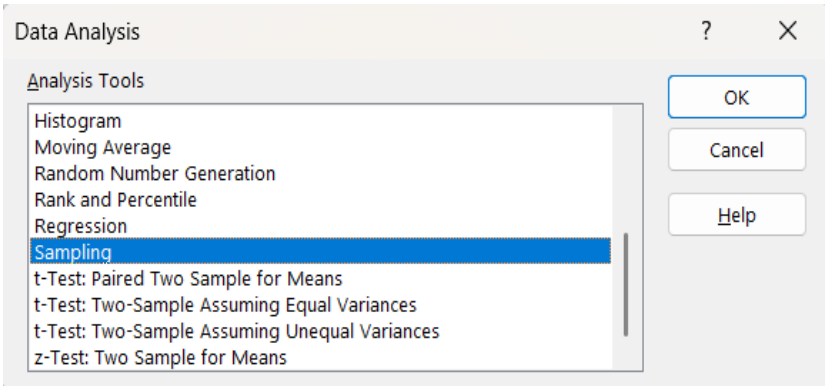
Example: From total students, 100 student's marks are given in the table below, create a periodic sample of 10 students.

You need to run the Sampling Data Analysis Tool to create a sample by selecting the 100 cells in the marks column as Input Range and periodic with period 10.

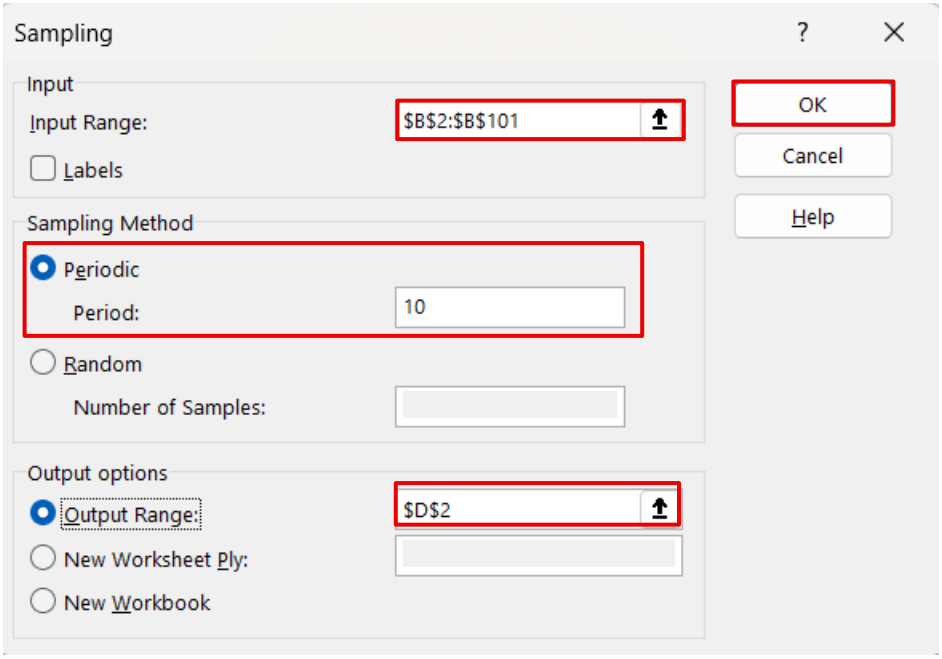
For Periodic Sampling:

To Perform this,

Go to Data Tab, Click Data Analysis, Scroll and Select Sampling and Click OK.



Then, Select the Input Range, Select Periodic Radio Button and type 10, then select the Output Range.



Output:

Roll no.	Marks		Periodic 10 Sample of Marks from total
1	46		25
2	29		21
3	34		6
4	2		32
5	31		1
6	37		41
7	6		46
8	44		28
9	3		35
10	25		7
11	41		
12	20		
13	16		
14	12		
15	15		
16	11		
17	44		
18	30		
19	5		
20	21		
21	3		
22	16		
23	44		
24	18		
25	22		
26	19		